

|                                                                                    |                                                                                                                       |                                                                                                                              |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>GENERIC REFINERY - GULF COAST</b><br>100 kBPSD Crude & Vacuum Distillation Unit | <b>MECHANICAL DESIGN CALCULATIONS</b><br>Columns, drums, desalters, fired-heater coils, nozzles, wind/skirts, weights | <b>CFU-000-ME-CAL-001</b><br>Rev A - Issued for review (FEED)<br>2026-10-02 - Prepared: CLAUDE CODE - Discipline: MECHANICAL |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|

## 1. Purpose and scope

FEED-level mechanical design of the CDU/VDU pressure equipment: shell/head/cone thickness to ASME VIII Div.1, external pressure (C-201 full vacuum) with stiffening rings, hydrotest pressures, empty / operating / hydrotest weights, skirt heights, ASCE 7 wind loads with skirt and anchor-bolt checks, API 530 fired-heater tube thickness, and main-column nozzle sizing. Inputs are read from data/equipment.json, streams.json, process\_results.json and psv.json (generator cfu/mech). Results are exported to data/mech.json for the layout, civil/foundation and cost-estimate work. Datasheets: CFU-000-ME-DS-001..007; GA drawings: CFU-100/200-ME-GA-001..008.

## 2. Design basis and assumptions

- Code: ASME BPVC VIII Div.1 (2023) - UG-16, UG-23, UG-27, UG-28, UG-29, UG-32, UG-33, UG-99; ASME II-D Table 1A allowable stresses (Section 3). Flanges ASME B16.5 / B16.47 Group 1.1 ratings.
- Design pressure / temperature and corrosion allowance from equipment.json. Static liquid head to HLL added to the design pressure of bottom courses and bottom heads (operating liquid density from H&MB;).
- Joint efficiency  $E = 1.0$  (full RT) for columns, desalters and drums  $\geq 2.0$  m ID;  $E = 0.85$  (spot RT) for smaller drums. Skirt-to-head weld efficiency 0.7.
- Fabrication minimum:  $t_{\min}$  (excl. CA) =  $\max(\text{UG-16(b)} \ 1.5 \text{ mm}, D/1000 + 2.5 \text{ mm})$ ; plus CA. Heads: 2:1 semi-ellipsoidal, nominal =  $\max(\text{min. after forming} + 1.5 \text{ mm thinning, adjoining shell})$ . Cones: 30 deg half-angle, UG-32(g); junction reinforcement per App. 1-5/1-8 by fabricator.
- Cladding / linings (410S, 317L, Monel 400) are corrosion barriers only - not credited for strength; included in weights (3 mm clad, 2 mm Monel lining).
- External pressure (C-201): 1.034 bar (15 psi) full vacuum at design temperature. Factor A from the Div.2 4.4.5.1 closed form (the basis of Fig. G), factor  $B = A.E/2$  in the elastic range with a plastic knee at  $0.5 S_y(T)$  representing chart CS-2. **Stiffening rings: external flat-bar rings at max. 3.0 m spacing** on all C-201 cylindrical courses (rings double as insulation/platform supports); cone-cylinder junctions taken as lines of support (to be confirmed by App. 1-8).
- Wind: ASCE 7-16 Ch.26/29,  $V = 150$  mph (67.1 m/s, 3-s gust), Exposure C,  $K_d = 0.95$ ,  $K_{zt} = 1.0$ ,  $C_f = 0.7$  (round, moderately smooth), effective diameter = insulated OD + 0.6 m (ladders, piping), platforms  $1.2 \text{ m}^2 \times C_f \ 2.0$  each. Gust factor  $G = 0.85$  for rigid structures ( $n_1 \geq 1 \text{ Hz}$ );  $G_f$  per 26.11.5 ( $\beta = 1 \%$ ) for flexible columns. Stress checks use ASD  $0.6W$  (+  $0.6D$  for anchor uplift).
- Longitudinal stress (UG-23): tension (windward, design pressure, empty weight)  $\leq S.E$ ; compression (leeward, operating weight, + vacuum for C-201)  $\leq B$  (UG-23(b),  $A = 0.125/(Ro/t)$ ). No 1.2 wind increase taken.
- Skirt height set by NPSH of the bottoms pump:  $LLL \geq \text{pump CL (EL 100.800)} + \text{NPSHr(est.)} + 1.0 \text{ m margin} + 0.6 \text{ m suction losses}$ , with minimum 3.0 m (5.0 m C-101, 6.0 m C-201). NPSHr estimated from suction-specific speed 200 (metric) - see datasheet CFU-000-ME-DS-006.
- Weights: steel density 7850 kg/m<sup>3</sup>; nozzles  $1.6 \text{ NPS}^3 \times 1.75 \text{ kg}$  (manways  $\times 1.4 + 150 \text{ kg}$ ); valve trays 60-75 kg/m<sup>2</sup> + support ring; structured packing 150-210 kg/m<sup>3</sup>, grid 380 kg/m<sup>3</sup>; distributors 120 kg/m<sup>2</sup>; collector trays 220 kg/m<sup>2</sup>; platforms 250 kg/m<sup>2</sup> (60 % wrap, 1.2 m wide) + ladders 35 kg/m; insulation mineral wool 130 kg/m<sup>3</sup> + 6 kg/m<sup>2</sup> cladding; skirt fireproofing 50 mm both faces. Operating liquid: tray clear liquid + downcomer backup, packing hold-up 5 %, sump to NLL. Hydrotest: vessel full of water (internals installed, no insulation).
- Accuracy: thickness to nominal plate; weights  $\pm 15 \%$  (vessels),  $\pm 30 \%$  (fired heaters, exchangers, air coolers). All values to be superseded by vendor/fabricator calculations.

## 3. Material allowable stresses (ASME II-D Table 1A)

| Spec        | Material                            | 40 C | 100 C | 150 C | 200 C | 250 C | 300 C | 325 C | 350 C | 375 C | 400 C | 425 C | 450 C | 475 C | 500 C |
|-------------|-------------------------------------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| SA-516-70   | SA-516 Gr.70 (CS plate, normalised) | 138  | 138   | 138   | 138   | 138   | 135   | 132   | 128   | 122   | 101   | 84    | 67    | 51    | 34    |
| SA-387-11-2 | SA-387 Gr.11 Cl.2 (1.25Cr-0.5Mo)    | 148  | 148   | 148   | 148   | 148   | 148   | 148   | 148   | 147   | 145   | 142   | 139   | 131   | 107   |
| SA-387-5-2  | SA-387 Gr.5 Cl.2 (5Cr-0.5Mo)        | 148  | 148   | 148   | 148   | 148   | 148   | 148   | 147   | 146   | 144   | 140   | 128   | 99    | 79    |

S in MPa; customary-unit values converted and interpolated. SA-516-70 above 425 C is subject to graphitisation limits (not used above 415 C here).

| Tag    | Base material | Design T (C) | S at design T (MPa) | S at test T (MPa) | E    | CA (mm) |
|--------|---------------|--------------|---------------------|-------------------|------|---------|
| C-101  | SA-516-70     | 390          | 109.3               | 137.9             | 1.0  | 3.0     |
| C-102  | SA-516-70     | 250          | 137.9               | 137.9             | 1.0  | 3.0     |
| C-103  | SA-516-70     | 330          | 131.6               | 137.9             | 1.0  | 6.0     |
| C-104  | SA-516-70     | 375          | 121.6               | 137.9             | 1.0  | 6.0     |
| C-105  | SA-516-70     | 235          | 137.9               | 137.9             | 1.0  | 3.0     |
| C-106  | SA-516-70     | 185          | 137.9               | 137.9             | 1.0  | 3.0     |
| C-201  | SA-516-70     | 415          | 90.8                | 137.9             | 1.0  | 6.0     |
| D-101A | SA-516-70     | 165          | 137.9               | 137.9             | 1.0  | 3.0     |
| D-101B | SA-516-70     | 165          | 137.9               | 137.9             | 1.0  | 3.0     |
| D-102  | SA-516-70     | 160          | 137.9               | 137.9             | 1.0  | 6.0     |
| D-103  | SA-516-70     | 90           | 137.9               | 137.9             | 0.85 | 3.0     |
| D-104  | SA-516-70     | 280          | 136.4               | 137.9             | 1.0  | 3.0     |
| D-105  | SA-516-70     | 95           | 137.9               | 137.9             | 0.85 | 3.0     |
| D-106  | SA-516-70     | 120          | 137.9               | 137.9             | 1.0  | 3.0     |
| D-201  | SA-516-70     | 100          | 137.9               | 137.9             | 0.85 | 3.0     |
| D-202  | SA-516-70     | 100          | 137.9               | 137.9             | 0.85 | 3.0     |

## 4. Formulae

- Shell, circumferential stress UG-27(c)(1):  $t = P.R / (S.E - 0.6 P)$ ; longitudinal UG-27(c)(2):  $t = P.R / (2 S.E + 0.4 P)$ ; R = inside radius in corroded condition.
- 2:1 ellipsoidal head UG-32(d):  $t = P.D / (2 S.E - 0.2 P)$ . Cone UG-32(g):  $t = P.D / (2 \cos(a) (S.E - 0.6 P))$ .
- External pressure UG-28(c):  $P_a = 4B / (3 Do/t)$ ; UG-33(d) ellipsoidal head:  $P_a = B / (Ro/t)$ ,  $Ro = 0.9 Do$ ; cone UG-33(f): equivalent cylinder  $Le = (L/2)(1 + Ds/DL)$ ,  $t_e = t \cos(a)$ .
- Stiffening ring UG-29:  $I_s = Do^2 L_s (t + As/Ls) A / 14$ , A from  $B = 0.75 P Do / (t + As/Ls)$ .
- Hydrotest UG-99(b):  $P_t = 1.3 MAWP (S_{test} / S_{design})$ , MAWP taken = design pressure (top); bottom adds water head. Membrane stress at test  $\leq 0.9 S_y$  (vertical field test).
- API 530: elastic  $t_{\sigma} = P_{el}.Do / (2 \sigma_{el} + P_{el})$ ,  $t = t_{\sigma} + CA$ ; rupture  $t_{\sigma} = P_r.Do / (2 \sigma_r + P_r)$ ,  $t = t_{\sigma} + f_{corr}.CA$  ( $f_{corr} = 0.6$ ).

## 5. Columns - shell, cone and head thickness (internal pressure)

**C-101 - Atmospheric crude fractionator (SA-516-70, P = 3.5 barg, T = 390 C, S = 109.3 MPa, E = 1.0, CA = 3.0 mm)**

| Component                       | ID (mm) | z (m from BTL) | P calc (barg) | t circ | t long | t min fab | CA  | t req (int) | t ext (FV) | t wind | t nominal (mm)       |
|---------------------------------|---------|----------------|---------------|--------|--------|-----------|-----|-------------|------------|--------|----------------------|
| Stripping section               | 4,140   | 0.00-10.02     | 3.74          | 7.11   | 3.55   | 6.6       | 3.0 | 10.1        | -          | 20     | 20                   |
| Swage 30 deg                    | 6,900   | 10.02-12.41    | 3.50          | 12.79  | -      | 9.4       | 3.0 | 15.8        | -          | 18     | 18                   |
| Flash zone / rectifying section | 6,900   | 12.41-40.50    | 3.50          | 11.08  | 5.53   | 9.4       | 3.0 | 14.1        | -          | -      | 16                   |
| Head (top) 2:1 SE               | 6,900   | -              | 3.50          | 11.06  | -      | -         | 3.0 | 14.1        | -          | -      | 16 (min 14.1 formed) |
| Head (bottom) 2:1 SE            | 4,140   | -              | 3.81          | 7.24   | -      | -         | 3.0 | 10.2        | -          | -      | 12 (min 10.2 formed) |

**C-102 - Kero side stripper (SA-516-70, P = 3.5 barg, T = 250 C, S = 137.9 MPa, E = 1.0, CA = 3.0 mm)**

| Component            | ID (mm) | z (m from BTL) | P calc (barg) | t circ | t long | t min fab | CA  | t req (int) | t ext (FV) | t wind | t nominal (mm)             |
|----------------------|---------|----------------|---------------|--------|--------|-----------|-----|-------------|------------|--------|----------------------------|
| Stripping section    | 1,700   | 0.00-9.20      | 3.65          | 2.26   | 1.13   | 4.2       | 3.0 | 7.2         | -          | -      | <b>8</b>                   |
| Head (top) 2:1 SE    | 1,700   | -              | 3.50          | 2.17   | -      | -         | 3.0 | 7.2         | -          | -      | <b>10</b> (min 7.2 formed) |
| Head (bottom) 2:1 SE | 1,700   | -              | 3.68          | 2.27   | -      | -         | 3.0 | 7.2         | -          | -      | <b>10</b> (min 7.2 formed) |

**C-103 - Diesel side stripper (SA-516-70, P = 3.5 barg, T = 330 C, S = 131.6 MPa, E = 1.0, CA = 6.0 mm)**

| Component            | ID (mm) | z (m from BTL) | P calc (barg) | t circ | t long | t min fab | CA  | t req (int) | t ext (FV) | t wind | t nominal (mm)              |
|----------------------|---------|----------------|---------------|--------|--------|-----------|-----|-------------|------------|--------|-----------------------------|
| Stripping section    | 1,900   | 0.00-9.20      | 3.65          | 2.65   | 1.32   | 4.4       | 6.0 | 10.4        | -          | -      | <b>12</b>                   |
| Head (top) 2:1 SE    | 1,900   | -              | 3.50          | 2.54   | -      | -         | 6.0 | 10.4        | -          | -      | <b>12</b> (min 10.4 formed) |
| Head (bottom) 2:1 SE | 1,900   | -              | 3.67          | 2.67   | -      | -         | 6.0 | 10.4        | -          | -      | <b>12</b> (min 10.4 formed) |

**C-104 - Ago side stripper (SA-516-70, P = 3.5 barg, T = 375 C, S = 121.6 MPa, E = 1.0, CA = 6.0 mm)**

| Component            | ID (mm) | z (m from BTL) | P calc (barg) | t circ | t long | t min fab | CA  | t req (int) | t ext (FV) | t wind | t nominal (mm)             |
|----------------------|---------|----------------|---------------|--------|--------|-----------|-----|-------------|------------|--------|----------------------------|
| Stripping section    | 1,400   | 0.00-9.20      | 3.66          | 2.13   | 1.06   | 3.9       | 6.0 | 9.9         | -          | -      | <b>10</b>                  |
| Head (top) 2:1 SE    | 1,400   | -              | 3.50          | 2.03   | -      | -         | 6.0 | 9.9         | -          | -      | <b>12</b> (min 9.9 formed) |
| Head (bottom) 2:1 SE | 1,400   | -              | 3.68          | 2.14   | -      | -         | 6.0 | 9.9         | -          | -      | <b>12</b> (min 9.9 formed) |

**C-105 - Naphtha stabiliser (debutaniser) (SA-516-70, P = 13.0 barg, T = 235 C, S = 137.9 MPa, E = 1.0, CA = 3.0 mm)**

| Component            | ID (mm) | z (m from BTL) | P calc (barg) | t circ | t long | t min fab | CA  | t req (int) | t ext (FV) | t wind | t nominal (mm)              |
|----------------------|---------|----------------|---------------|--------|--------|-----------|-----|-------------|------------|--------|-----------------------------|
| Shell                | 2,100   | 0.00-18.00     | 13.13         | 10.08  | 5.00   | 4.6       | 3.0 | 13.1        | -          | -      | <b>14</b>                   |
| Head (top) 2:1 SE    | 2,100   | -              | 13.00         | 9.94   | -      | -         | 3.0 | 12.9        | -          | -      | <b>16</b> (min 12.9 formed) |
| Head (bottom) 2:1 SE | 2,100   | -              | 13.16         | 10.06  | -      | -         | 3.0 | 13.1        | -          | -      | <b>16</b> (min 13.1 formed) |

**C-106 - Naphtha splitter (SA-516-70, P = 3.5 barg, T = 185 C, S = 137.9 MPa, E = 1.0, CA = 3.0 mm)**

| Component            | ID (mm) | z (m from BTL) | P calc (barg) | t circ | t long | t min fab | CA  | t req (int) | t ext (FV) | t wind | t nominal (mm)             |
|----------------------|---------|----------------|---------------|--------|--------|-----------|-----|-------------|------------|--------|----------------------------|
| Shell                | 3,100   | 0.00-29.50     | 3.65          | 4.11   | 2.05   | 5.6       | 3.0 | 8.6         | -          | 12     | <b>12</b>                  |
| Head (top) 2:1 SE    | 3,100   | -              | 3.50          | 3.94   | -      | -         | 3.0 | 8.6         | -          | -      | <b>12</b> (min 8.6 formed) |
| Head (bottom) 2:1 SE | 3,100   | -              | 3.69          | 4.16   | -      | -         | 3.0 | 8.6         | -          | -      | <b>12</b> (min 8.6 formed) |

**C-201 - Vacuum column (wet, packed) (SA-516-70, P = 3.5 barg + FV, T = 415 C, S = 90.8 MPa, E = 1.0, CA = 6.0 mm)**

| Component                         | ID (mm) | z (m from BTL) | P calc (barg) | t circ | t long | t min fab | CA  | t req (int) | t ext (FV) | t wind | t nominal (mm)              |
|-----------------------------------|---------|----------------|---------------|--------|--------|-----------|-----|-------------|------------|--------|-----------------------------|
| Boot (stripping / quench)         | 3,000   | 0.00-7.77      | 3.75          | 6.24   | 3.11   | 5.5       | 6.0 | 12.2        | 18         | 25     | <b>25</b>                   |
| Lower cone 30 deg                 | 6,600   | 7.77-10.89     | 3.50          | 14.75  | -      | 9.1       | 6.0 | 20.7        | 25         | -      | <b>25</b>                   |
| Main shell (beds 2-4, flash zone) | 6,600   | 10.89-27.04    | 3.50          | 12.77  | 6.37   | 9.1       | 6.0 | 18.8        | 24         | -      | <b>25</b>                   |
| Upper cone 30 deg                 | 6,600   | 27.04-30.50    | 3.50          | 14.75  | -      | 9.1       | 6.0 | 20.7        | 25         | -      | <b>25</b>                   |
| Top section (bed 1)               | 2,600   | 30.50-36.00    | 3.50          | 5.04   | 2.51   | 5.1       | 6.0 | 11.1        | 17         | -      | <b>18</b>                   |
| Head (top) 2:1 SE                 | 2,600   | -              | 3.50          | 5.03   | -      | -         | 6.0 | 11.1        | 14         | -      | <b>18</b> (min 14.0 formed) |
| Head (bottom) 2:1 SE              | 3,000   | -              | 3.81          | 6.32   | -      | -         | 6.0 | 12.3        | 15         | -      | <b>18</b> (min 15.0 formed) |

**6. C-201 external pressure (full vacuum) and stiffening rings**

| Course                            | Do (mm) | t nom | t corr/eff | L (mm) | L/Do  | Do/t | A        | B (MPa) | Pa (bar) | Req. (bar) | t if no rings |
|-----------------------------------|---------|-------|------------|--------|-------|------|----------|---------|----------|------------|---------------|
| Boot (stripping / quench)         | 3,036   | 18    | 12.0       | 3,000  | 0.988 | 253  | 2.66e-04 | 23.2    | 1.222    | 1.034      | 24            |
| Lower cone 30 deg                 | 6,650   | 25    | 16.5       | 2,267  | 0.341 | 404  | 4.00e-04 | 34.6    | 1.141    | 1.034      | -             |
| Main shell (beds 2-4, flash zone) | 6,648   | 24    | 18.0       | 3,000  | 0.451 | 369  | 3.41e-04 | 29.6    | 1.070    | 1.034      | 43            |
| Upper cone 30 deg                 | 6,650   | 25    | 16.5       | 2,414  | 0.363 | 404  | 3.74e-04 | 32.4    | 1.070    | 1.034      | -             |
| Top section (bed 1)               | 2,634   | 17    | 11.0       | 3,000  | 1.139 | 239  | 2.49e-04 | 21.7    | 1.209    | 1.034      | 20            |

Design temperature 415 C, material SA-516-70, CA 6.0 mm; cones use the equivalent-cylinder method. Without rings the main shell would need the thickness in the last column.

| Course                            | No. of rings | Spacing Ls (mm) | Flat bar h x b (mm) | As (mm2) | A        | B (MPa) | Is req (10^6 mm4) | I provided (10^6 mm4) |
|-----------------------------------|--------------|-----------------|---------------------|----------|----------|---------|-------------------|-----------------------|
| Boot (stripping / quench)         | 2            | 3,000           | 160 x 16            | 2,560    | 2.10e-04 | 18.3    | 5.33              | 5.46                  |
| Main shell (beds 2-4, flash zone) | 5            | 3,000           | 300 x 25            | 7,500    | 2.88e-04 | 25.1    | 55.96             | 56.25                 |
| Top section (bed 1)               | 1            | 3,000           | 140 x 16            | 2,240    | 1.99e-04 | 17.4    | 3.48              | 3.66                  |

External rings, continuous fillet welds both sides (UG-30). Ring stiffeners also carry insulation support. Internal bed-support / collector rings are not credited.

Heads under external pressure (UG-33(d)): top head t = 14 mm, bottom head t = 15 mm minimum (corroded 6.0 mm deducted); nominal 18 / 18 mm.

## 7. Hydrotest

| Tag    | MAWP (barg) | S_test/S_des | Pt top (barg) | Pt bottom (barg) | sigma bottom (MPa) | 0.9 Sy (MPa) | Check | Test position     |
|--------|-------------|--------------|---------------|------------------|--------------------|--------------|-------|-------------------|
| C-101  | 3.5         | 1.262        | 5.74          | 9.98             | 104                | 234          | OK    | Vertical (field)  |
| C-102  | 3.5         | 1.000        | 4.55          | 5.54             | 59                 | 234          | OK    | Vertical (field)  |
| C-103  | 3.5         | 1.048        | 4.77          | 5.76             | 46                 | 234          | OK    | Vertical (field)  |
| C-104  | 3.5         | 1.134        | 5.16          | 6.13             | 43                 | 234          | OK    | Vertical (field)  |
| C-105  | 13.0        | 1.000        | 16.90         | 18.77            | 142                | 234          | OK    | Vertical (field)  |
| C-106  | 3.5         | 1.000        | 4.55          | 7.60             | 99                 | 234          | OK    | Vertical (field)  |
| C-201  | 3.5         | 1.518        | 6.91          | 10.58            | 64                 | 234          | OK    | Vertical (field)  |
| D-101A | 17.0        | 1.000        | 22.10         | 22.47            | -                  | -            | OK    | Horizontal (shop) |
| D-101B | 17.0        | 1.000        | 22.10         | 22.47            | -                  | -            | OK    | Horizontal (shop) |
| D-102  | 3.5         | 1.000        | 4.55          | 4.88             | -                  | -            | OK    | Horizontal (shop) |
| D-103  | 7.0         | 1.000        | 9.10          | 9.19             | -                  | -            | OK    | Vertical          |
| D-104  | 3.5         | 1.011        | 4.60          | 4.81             | -                  | -            | OK    | Horizontal (shop) |
| D-105  | 15.2        | 1.000        | 19.76         | 19.93            | -                  | -            | OK    | Horizontal (shop) |
| D-106  | 3.5         | 1.000        | 4.55          | 4.81             | -                  | -            | OK    | Horizontal (shop) |
| D-201  | 3.5         | 1.000        | 4.55          | 4.68             | -                  | -            | OK    | Horizontal (shop) |
| D-202  | 3.5         | 1.000        | 4.55          | 4.61             | -                  | -            | OK    | Vertical          |

Foundations for C-101 / C-201 shall be checked for the full-of-water hydrotest weight (Section 8) unless a pneumatic/hydro-pneumatic or horizontal shop test is specified.

## 8. Weights

| Tag   | Shell + heads | Skirt + base | Nozzles | Internals | Clad | Platf./ladders | Insul./fireproof. | EMPTY | Op. liquid | OPERATING | Volume (m3) | HYDROTEST |
|-------|---------------|--------------|---------|-----------|------|----------------|-------------------|-------|------------|-----------|-------------|-----------|
| C-101 | 115.0         | 17.0         | 8.8     | 119.0     | 15.7 | 38.9           | 38.4              | 352.8 | 85.6       | 438.3     | 1,296       | 1629.1    |
| C-102 | 3.7           | 2.8          | 1.8     | 1.4       | 0.0  | 4.6            | 5.7               | 19.9  | 2.9        | 22.9      | 22          | 41.2      |
| C-103 | 6.1           | 3.4          | 1.8     | 1.6       | 1.4  | 4.9            | 7.3               | 26.5  | 3.6        | 30.1      | 28          | 53.1      |
| C-104 | 3.7           | 2.2          | 1.7     | 1.1       | 1.0  | 4.3            | 4.4               | 18.4  | 2.1        | 20.4      | 15          | 32.3      |
| C-105 | 14.7          | 4.4          | 2.8     | 5.3       | 0.0  | 9.9            | 11.0              | 48.1  | 5.8        | 53.9      | 65          | 110.7     |
| C-106 | 29.8          | 9.6          | 4.7     | 24.8      | 0.0  | 18.3           | 21.3              | 108.7 | 21.0       | 129.7     | 230         | 334.1     |
| C-201 | 113.8         | 10.8         | 8.8     | 97.9      | 12.8 | 24.0           | 25.9              | 294.0 | 43.2       | 337.1     | 763         | 1043.4    |

Tonnes. EMPTY = installed, incl. internals, platforms, insulation and fireproofing (erection lift excludes platforms/insulation if installed after setting).

## Drums and desalters - thickness and weights

| Tag    | Service                                | ID x T/T (m) | Material  | P calc (barg) | Shell t req | Shell t nom | Head t req | Head t nom | Fabricated (t) | Empty (t) | Operating (t) | Hydrotest (t) |
|--------|----------------------------------------|--------------|-----------|---------------|-------------|-------------|------------|------------|----------------|-----------|---------------|---------------|
| D-101A | Electrostatic desalter, 1st stage      | 3.8 x 30.0   | SA-516-70 | 17.29         | 27.0        | 28          | 26.9       | 28         | 119.0          | 137.4     | 414.0         | 487.1         |
| D-101B | Electrostatic desalter, 2nd stage      | 3.8 x 30.0   | SA-516-70 | 17.29         | 27.0        | 28          | 26.9       | 28         | 119.0          | 137.4     | 414.0         | 487.1         |
| D-102  | Atm. overhead reflux drum (3-phase)    | 3.4 x 12.0   | SA-516-70 | 3.73          | 11.9        | 12          | 11.9       | 12         | 19.4           | 19.4      | 63.0          | 144.0         |
| D-103  | Fuel gas knock-out drum                | 0.9 x 2.5    | SA-516-70 | 7.17          | 6.4         | 8           | 6.4        | 8          | 1.2            | 1.2       | 1.8           | 3.0           |
| D-104  | Flare knock-out drum (unit)            | 2.1 x 6.5    | SA-516-70 | 3.64          | 7.6         | 8           | 7.6        | 8          | 4.5            | 5.2       | 13.9          | 29.5          |
| D-105  | Stabiliser reflux drum                 | 1.7 x 5.5    | SA-516-70 | 15.32         | 14.2        | 16          | 14.2       | 16         | 6.2            | 6.2       | 11.2          | 20.4          |
| D-106  | Splitter reflux drum                   | 2.7 x 8.5    | SA-516-70 | 3.69          | 8.2         | 10          | 8.2        | 10         | 8.9            | 8.9       | 27.8          | 62.8          |
| D-201  | Ejector hotwell / sour water separator | 1.3 x 5.5    | SA-516-70 | 3.63          | 6.8         | 8           | 6.8        | 8          | 2.5            | 2.5       | 6.4           | 10.4          |
| D-202  | Vacuum off-gas knock-out drum          | 0.6 x 2.0    | SA-516-70 | 3.64          | 6.1         | 8           | 6.1        | 8          | 0.9            | 0.9       | 1.1           | 1.5           |

Desalters operate liquid-full (crude/water, 780 kg/m<sup>3</sup>); include 3 transformers (4.5 t each) and 18 t internals. Drums at 50 % liquid. Boots included (D-102, D-105).

## 9. Skirt height, wind load and longitudinal stress

| Tag   | Bottoms pump | NPSHr est. (m) | LLL above BTL (m) | Skirt height (m) | BTL elevation | Skirt t (mm) | Skirt OD (mm) |
|-------|--------------|----------------|-------------------|------------------|---------------|--------------|---------------|
| C-101 | P-112A/B     | 4.2            | 0.6               | 6.5              | 106.500       | 20           | 4,180         |
| C-102 | P-109A/B     | 1.9            | 0.5               | 4.0              | 104.000       | 8            | 1,716         |
| C-103 | P-110A/B     | 2.2            | 0.5               | 4.5              | 104.500       | 8            | 1,924         |
| C-104 | P-111A/B     | 1.5            | 0.5               | 3.5              | 103.500       | 8            | 1,420         |
| C-105 | - (no pump)  | -              | 0.5               | 6.0              | 106.000       | 8            | 2,128         |
| C-106 | P-117A/B     | 5.4            | 0.5               | 7.5              | 107.500       | 12           | 3,124         |
| C-201 | P-204A/B     | 2.6            | 0.6               | 6.0              | 106.000       | 18           | 3,050         |

C-105 bottoms flow under pressure to C-106 / kettle E-116; skirt 6.0 m for kettle liquid head and bottom piping.

| Tag   | Height (m) | n1 (Hz) | Type     | G / Gf | Kz top | qh (kPa) | Base shear (kN) | Base moment (kNm) | ASD moment (kNm) | Vortex Vcr (m/s) |
|-------|------------|---------|----------|--------|--------|----------|-----------------|-------------------|------------------|------------------|
| C-101 | 48.7       | 0.83    | flexible | 1.113  | 1.397  | 3.66     | 863             | 24,853            | 14,912           | 29.7             |
| C-102 | 13.6       | 4.25    | rigid    | 0.890  | 1.068  | 2.80     | 60              | 485               | 291              | 39.3             |
| C-103 | 14.2       | 4.26    | rigid    | 0.889  | 1.077  | 2.82     | 67              | 562               | 337              | 44.8             |
| C-104 | 13.0       | 3.59    | rigid    | 0.890  | 1.059  | 2.77     | 53              | 416               | 250              | 28.7             |
| C-105 | 24.5       | 1.66    | rigid    | 0.882  | 1.209  | 3.17     | 138             | 1,992             | 1,195            | 18.6             |
| C-106 | 37.8       | 1.16    | rigid    | 0.876  | 1.324  | 3.47     | 294             | 6,308             | 3,785            | 18.9             |
| C-201 | 42.6       | 0.81    | flexible | 1.223  | 1.358  | 3.56     | 650             | 15,673            | 9,404            | 11.6             |

Strength-level wind. Vortex shedding: critical velocity  $V_{cr} = n1.D/0.2$ . Columns with  $V_{cr} < \sim 25$  m/s and  $D/t > 200$  to be checked in detail (helical strakes not expected for insulated columns with platforms).

### C-101 - wind / weight longitudinal stress check (corroded, ASD)

| Section (bottom of)              | Elev. above grade (m) | t nom | M ASD (MNm) | sig P | sig M | sig W | Tension | Allow. S.E | Compression | Allow. B | Result |
|----------------------------------|-----------------------|-------|-------------|-------|-------|-------|---------|------------|-------------|----------|--------|
| Stripping section                | 6.50                  | 20    | 12          | 21.3  | 50.4  | 17.7  | 57.9    | 109.3      | 68.1        | 73.3     | OK     |
| Swage 30 deg                     | 16.52                 | 18    | 7           | 27.9  | 40.3  | 19.1  | 52.3    | 109.3      | 59.5        | 62.9     | OK     |
| Flash zone / rectifying section  | 18.91                 | 16    | 6           | 46.4  | 12.5  | 11.0  | 49.9    | 109.3      | 23.6        | 40.9     | OK     |
| Skirt (base, 1.5 mm CA, E = 0.7) | 0.00                  | 20    | 15          | -     | 58.2  | 16.9  | 44.8    | 96.5       | 75.1        | 93.5     | OK     |

Anchor bolts C-101: 32 x M56 ASTM F1554 Gr.55 on 4.43 m BCD; max. tension 363 kN/bolt (0.6D + 0.6W, empty), required root area 1910 mm<sup>2</sup>. Hydrotest skirt compression 80.0 MPa (25 % wind). Base shear 863 kN, base moment 24853 kNm (strength) for foundation design.

### C-201 - wind / weight longitudinal stress check (corroded, ASD)

| Section (bottom of)               | Elev. above grade (m) | t nom | M ASD (MNm) | sig P | sig M | sig W | Tension | Allow. S.E | Compression | Allow. B | Result |
|-----------------------------------|-----------------------|-------|-------------|-------|-------|-------|---------|------------|-------------|----------|--------|
| Boot (stripping / quench)         | 6.00                  | 25    | 7           | 13.8  | 52.4  | 17.0  | 51.6    | 90.8       | 73.5        | 80.2     | OK     |
| Lower cone 30 deg                 | 13.77                 | 25    | 4           | 16.0  | 37.8  | 17.2  | 38.4    | 90.8       | 59.7        | 78.4     | OK     |
| Main shell (beds 2-4, flash zone) | 16.89                 | 25    | 3           | 30.4  | 5.3   | 6.4   | 30.1    | 90.8       | 20.7        | 58.1     | OK     |
| Upper cone 30 deg                 | 33.04                 | 25    | 0           | 35.1  | 0.7   | 1.0   | 34.8    | 90.8       | 12.1        | 51.8     | OK     |
| Top section (bed 1)               | 36.50                 | 18    | 0           | 19.0  | 2.6   | 2.3   | 19.4    | 90.8       | 10.5        | 75.1     | OK     |
| Skirt (base, 1.5 mm CA, E = 0.7)  | 0.00                  | 18    | 9           | -     | 77.2  | 20.0  | 59.8    | 96.5       | 97.2        | 102.2    | OK     |

Anchor bolts C-201: 24 x M64 ASTM F1554 Gr.55 on 3.30 m BCD; max. tension 409 kN/bolt (0.6D + 0.6W, empty), required root area 2154 mm<sup>2</sup>. Hydrotest skirt compression 83.7 MPa (25 % wind). Base shear 650 kN, base moment 15673 kNm (strength) for foundation design.

## 10. Fired heater tubes - API 530 (9Cr-1Mo, 100,000 h)

| Item                                                   | H-101 radiant            | H-201 radiant            |
|--------------------------------------------------------|--------------------------|--------------------------|
| Radiant average flux (kW/m <sup>2</sup> , OD)          | 31.5                     | 25.0                     |
| Peak flux (x1.8 circ. x1.1 long.) (kW/m <sup>2</sup> ) | 62.4                     | 49.5                     |
| Inside peak flux (kW/m <sup>2</sup> )                  | 71.7                     | 56.9                     |
| Bulk fluid T at outlet (C)                             | 365.7                    | 398.4                    |
| Inside film coefficient (W/m <sup>2</sup> K, EOR)      | 1,100                    | 650                      |
| Coke/fouling resistance EOR (m <sup>2</sup> K/W)       | 0.0005                   | 0.0009                   |
| dT film (C)                                            | 65.2                     | 87.6                     |
| dT coke (C)                                            | 35.9                     | 51.2                     |
| dT wall (C)                                            | 27.2                     | 21.6                     |
| Estimated max. TMT EOR (C)                             | 494                      | 559                      |
| <b>Design metal temperature (TMT + 15 C)</b>           | 510                      | 575                      |
| equipment.json des_T (C)                               | 455                      | 490                      |
| Elastic design pressure (MPa)                          | 3.85                     | 1.98                     |
| Rupture design pressure (MPa, coil inlet op.)          | 1.70                     | 1.90                     |
| sigma_el at Tdm (MPa)                                  | 101.8                    | 78.0                     |
| sigma_r 100,000 h at Tdm (MPa)                         | 83.0                     | 35.0                     |
| Corrosion allowance (mm)                               | 3.0                      | 3.0                      |
| Elastic t_sigma (mm)                                   | 3.12                     | 2.11                     |
| Rupture t_sigma (mm)                                   | 1.71                     | 4.45                     |
| Elastic t_min = t_sigma + CA                           | 6.12                     | 5.11                     |
| Rupture t_min = t_sigma + f_corr CA                    | 3.51                     | 6.25                     |
| <b>Minimum thickness (mm)</b>                          | 6.12                     | 6.25                     |
| Governing                                              | elastic                  | rupture                  |
| Minimum schedule                                       | Sch 40 (7.11 mm)         | Sch 40 (7.11 mm)         |
| <b>Selected</b> (OD 168.3 mm, A335 P9)                 | <b>Sch 80 (10.97 mm)</b> | <b>Sch 80 (10.97 mm)</b> |

Sch 80 is retained (process mass-flux basis ID 146.3 mm) - gives margin for coke spalling/erosion and decoking. 9Cr-1Mo stresses digitised from API 530 curves: 425 C: 123/205, 450 C: 118/160, 475 C: 112/123, 500 C: 105/93, 525 C: 97/68, 550 C: 88/49, 575 C: 78/35, 600 C: 67/25 MPa (sigma\_el/sigma\_r).

| Heater | Fired (MW) | APH duty (MW) | Radiant (MW) | Flue gas (t/h) | BWT (C) | Crossover (C) | Flue out conv. (C) | LMTD (C) | Conv. bare area (m2) | Rows (shock+studded+coils) | Utility coil req. (MW) |
|--------|------------|---------------|--------------|----------------|---------|---------------|--------------------|----------|----------------------|----------------------------|------------------------|
| H-101  | 67.4       | 6.4           | 38.7         | 96             | 1,044   | 306           | 381                | 327      | 607                  | 2 + 6 + 2 SS + 0 util.     | 0.00                   |
| H-201  | 16.7       | 0.0           | 9.5          | 24             | 854     | 366           | 399                | 192      | 185                  | 2 + 3 + 0 SS + 2 util.     | 1.40                   |

Convection estimate:  $U = 105 \text{ W/m}^2\text{K}$  on bare-tube area (studded), 50 C minimum approach to process inlet. Vendor to optimise.

## 11. Nozzle sizing summary (main columns)

### C-101 - Atmospheric crude fractionator

| Mark   | Service                                                   | Size   | Rating | z from BTL (m) | Flow (m3/h act.) | rho (kg/m3) | v (m/s) | Criterion                                                         |
|--------|-----------------------------------------------------------|--------|--------|----------------|------------------|-------------|---------|-------------------------------------------------------------------|
| N1     | Feed - transfer line from H-101 (tangential, vapour horn) | 36"    | 150#   | 14.71          | 34,016           | 17          | 15.0    | two-phase feed $\rho_{m.v2} \leq 6000 \text{ Pa}$                 |
| N2     | Overhead vapour to A-101                                  | 36"    | 150#   | 40.50          | 45,065           | 5.09        | 19.9    | vapour outlet $\rho_{m.v2} \leq 3000 \text{ Pa}$                  |
| N3     | Reflux from P-103                                         | 8"     | 150#   | 38.85          | 145              | 698         | 1.2     | reflux / pumped liquid $\leq 2.0 \text{ m/s}$                     |
| N4     | TPA return                                                | 8"     | 150#   | 38.80          | 246              | 679         | 2.1     | pumped liquid return $\leq 2.5 \text{ m/s}$                       |
| N5     | TPA draw to P-106 (draw sump)                             | 14"    | 150#   | 36.83          | 270              | 617         | 0.8     | gravity draw / pump suction $\leq 0.9 \text{ m/s}$ (self-venting) |
| N8     | MPA return                                                | 10"    | 150#   | 32.25          | 320              | 650         | 1.7     | pumped liquid return $\leq 2.5 \text{ m/s}$                       |
| N9     | MPA draw to P-107 (draw sump)                             | 16"    | 150#   | 30.28          | 352              | 591         | 0.8     | gravity draw / pump suction $\leq 0.9 \text{ m/s}$ (self-venting) |
| N12    | BPA return                                                | 8"     | 150#   | 24.48          | 263              | 630         | 2.3     | pumped liquid return $\leq 2.5 \text{ m/s}$                       |
| N13    | BPA draw to P-108 (draw sump)                             | 16"    | 150#   | 22.51          | 289              | 573         | 0.7     | gravity draw / pump suction $\leq 0.9 \text{ m/s}$ (self-venting) |
| N6     | Kero draw to C-102                                        | 12"    | 150#   | 32.41          | 128              | 591         | 0.5     | gravity draw to side stripper $\leq 0.6 \text{ m/s}$              |
| N7     | Vapour return from C-102                                  | 8"     | 150#   | 32.92          | 2,250            | 3.70        | 19.4    | vapour return $\rho_{m.v2} \leq 4000 \text{ Pa}$                  |
| N10    | Diesel draw to C-103                                      | 12"    | 150#   | 24.64          | 157              | 573         | 0.6     | gravity draw to side stripper $\leq 0.6 \text{ m/s}$              |
| N11    | Vapour return from C-103                                  | 8"     | 150#   | 25.15          | 2,621            | 3.75        | 22.6    | vapour return $\rho_{m.v2} \leq 4000 \text{ Pa}$                  |
| N14    | Ago draw to C-104                                         | 10"    | 150#   | 18.09          | 79               | 636         | 0.4     | gravity draw to side stripper $\leq 0.6 \text{ m/s}$              |
| N15    | Vapour return from C-104                                  | 6"     | 150#   | 18.60          | 1,382            | 3.97        | 20.6    | vapour return $\rho_{m.v2} \leq 4000 \text{ Pa}$                  |
| N16    | Stripping steam (from H-101 SS coil)                      | 10"    | 150#   | 5.77           | 4,886            | 1.56        | 26.7    | stripping steam $\leq 35 \text{ m/s}$                             |
| N17    | Atm. residue to P-112                                     | 16"    | 150#   | -1.03          | 365              | 704         | 0.9     | pump suction (bottoms) $\leq 1.0 \text{ m/s}$                     |
| N18    | Vent / steam-out                                          | 3"     | 150#   | 40.50          | -                | -           | -       | -                                                                 |
| N19    | PSV-1001 relief (4 x T orifice)                           | 4 x 8" | 150#   | 40.50          | -                | -           | -       | -                                                                 |
| N20A/B | LT / LG bridle (bottom sump)                              | 2 x 3" | 150#   | 0.30           | -                | -           | -       | -                                                                 |
| N21    | Wash-zone overflash / slop connection (spare)             | 4"     | 150#   | 16.26          | -                | -           | -       | -                                                                 |
| M1     | Manway 24" (davit)                                        | 24"    | 150#   | 39.50          | -                | -           | -       | -                                                                 |
| M2     | Manway 24" (davit)                                        | 24"    | 150#   | 33.48          | -                | -           | -       | -                                                                 |
| M3     | Manway 24" (davit)                                        | 24"    | 150#   | 27.54          | -                | -           | -       | -                                                                 |
| M4     | Manway 24" (davit)                                        | 24"    | 150#   | 20.38          | -                | -           | -       | -                                                                 |
| M5     | Manway 24" (davit)                                        | 24"    | 150#   | 13.31          | -                | -           | -       | -                                                                 |
| M6     | Manway 24" (davit)                                        | 24"    | 150#   | 7.89           | -                | -           | -       | -                                                                 |
| M7     | Manway 24" (davit)                                        | 24"    | 150#   | 5.02           | -                | -           | -       | -                                                                 |

### C-201 - Vacuum column (wet, packed)

| Mark | Service                                                     | Size | Rating | z from BTL (m) | Flow (m3/h act.) | rho (kg/m3) | v (m/s) | Criterion                                                                        |
|------|-------------------------------------------------------------|------|--------|----------------|------------------|-------------|---------|----------------------------------------------------------------------------------|
| N1   | Feed - transfer line from H-201 (tangential to vapour horn) | 66"  | 150#   | 12.89          | 464,525          | 0.56        | 59.8    | vacuum transfer line $\leq 70 \text{ m/s}$ ( $\rho_{m.v2} \ll 6000 \text{ Pa}$ ) |
| N2   | Overhead vapour to J-201 / E-202                            | 48"  | 150#   | 36.00          | 229,101          | 0.01        | 56.3    | vacuum vapour $\leq 60 \text{ m/s}$                                              |
| N3   | LVGO PA return (+ LVGO reflux)                              | 6"   | 150#   | 35.30          | 96               | 860         | 1.4     | pumped liquid return $\leq 2.5 \text{ m/s}$                                      |
| N4   | LVGO total draw to P-201                                    | 12"  | 150#   | 30.35          | 226              | 739         | 0.9     | gravity draw / pump suction $\leq 0.9 \text{ m/s}$ (self-venting)                |

| Mark   | Service                              | Size   | Rating | z from BTL (m) | Flow (m3/h act.) | rho (kg/m3) | v (m/s) | Criterion                                             |
|--------|--------------------------------------|--------|--------|----------------|------------------|-------------|---------|-------------------------------------------------------|
| N5     | LVGO internal reflux to bed 2        | 4"     | 150#   | 26.89          | 45               | 760         | 1.5     | pumped liquid return <= 2.5 m/s                       |
| N6     | HVGO total draw to P-202             | 16"    | 150#   | 22.09          | 374              | 708         | 0.9     | gravity draw / pump suction <= 0.9 m/s (self-venting) |
| N7     | HVGO PA return                       | 8"     | 150#   | 22.09          | 204              | 790         | 1.8     | pumped liquid return <= 2.5 m/s                       |
| N8     | Wash oil (HVGO) to bed 4             | 2"     | 150#   | 17.19          | 16               | 790         | 2.1     | pumped liquid return <= 2.5 m/s                       |
| N9     | Slop wax draw to P-203               | 3"     | 150#   | 14.34          | 11               | 714         | 0.6     | gravity draw / pump suction <= 0.9 m/s (self-venting) |
| N10    | Stripping steam                      | 6"     | 150#   | 4.74           | 1,498            | 1.86        | 22.3    | stripping steam <= 35 m/s                             |
| N11    | VR quench return (from E-201 outlet) | 3"     | 150#   | 3.75           | 30               | 905         | 1.7     | pumped liquid return <= 2.5 m/s                       |
| N12    | Vacuum residue to P-204              | 10"    | 150#   | -0.75          | 179              | 757         | 1.0     | pump suction (bottoms) <= 1.0 m/s                     |
| N13    | PSV-2001 relief (R)                  | 6"     | 150#   | 36.00          | -                | -           | -       | -                                                     |
| N14A/B | LT / LG bridle (boot)                | 2 x 3" | 150#   | 0.30           | -                | -           | -       | -                                                     |
| N15    | Vent / steam-out / N2 purge          | 4"     | 150#   | 36.00          | -                | -           | -       | -                                                     |
| M1     | Manway 24"                           | 24"    | 150#   | 31.55          | -                | -           | -       | -                                                     |
| M2     | Manway 24"                           | 24"    | 150#   | 23.29          | -                | -           | -       | -                                                     |
| M3     | Manway 24"                           | 24"    | 150#   | 17.49          | -                | -           | -       | -                                                     |
| M4     | Manway 24" (36" in flash zone)       | 36"    | 150#   | 11.69          | -                | -           | -       | -                                                     |
| M5     | Manway 24"                           | 24"    | 150#   | 4.40           | -                | -           | -       | -                                                     |

### C-102 - Kero side stripper

| Mark  | Service                              | Size   | Rating | z from BTL (m) | Flow (m3/h act.) | rho (kg/m3) | v (m/s) | Criterion                                |
|-------|--------------------------------------|--------|--------|----------------|------------------|-------------|---------|------------------------------------------|
| N1    | Kero feed from C-101 tray 10         | 12"    | 150#   | 8.00           | 128              | 591         | 0.5     | gravity draw to side stripper <= 0.6 m/s |
| N2    | Vapour return to C-101               | 8"     | 150#   | 9.20           | 2,250            | 3.70        | 19.4    | vapour return rho.v2 <= 4000 Pa          |
| N3    | Stripping steam                      | 4"     | 150#   | 4.05           | 944              | 1.56        | 31.9    | stripping steam <= 35 m/s                |
| N4    | Bottoms to P-109                     | 8"     | 150#   | -0.42          | 114              | 603         | 1.0     | pump suction (bottoms) <= 1.0 m/s        |
| N5A/B | LT / LG bridle                       | 2 x 2" | 150#   | 0.20           | -                | -           | -       | -                                        |
| N6    | Vent / steam-out / PSV-1009 (common) | 3"     | 150#   | 9.20           | -                | -           | -       | -                                        |
| M1    | Manway 24"                           | 24"    | 150#   | 8.45           | -                | -           | -       | -                                        |
| M2    | Manway 24"                           | 24"    | 150#   | 3.30           | -                | -           | -       | -                                        |

### C-103 - Diesel side stripper

| Mark  | Service                              | Size   | Rating | z from BTL (m) | Flow (m3/h act.) | rho (kg/m3) | v (m/s) | Criterion                                |
|-------|--------------------------------------|--------|--------|----------------|------------------|-------------|---------|------------------------------------------|
| N1    | Diesel feed from C-101 tray 22       | 12"    | 150#   | 8.00           | 157              | 573         | 0.6     | gravity draw to side stripper <= 0.6 m/s |
| N2    | Vapour return to C-101               | 8"     | 150#   | 9.20           | 2,621            | 3.75        | 22.6    | vapour return rho.v2 <= 4000 Pa          |
| N3    | Stripping steam                      | 6"     | 150#   | 4.05           | 1,066            | 1.56        | 15.9    | stripping steam <= 35 m/s                |
| N4    | Bottoms to P-110                     | 10"    | 150#   | -0.47          | 140              | 584         | 0.8     | pump suction (bottoms) <= 1.0 m/s        |
| N5A/B | LT / LG bridle                       | 2 x 2" | 150#   | 0.20           | -                | -           | -       | -                                        |
| N6    | Vent / steam-out / PSV-1009 (common) | 3"     | 150#   | 9.20           | -                | -           | -       | -                                        |
| M1    | Manway 24"                           | 24"    | 150#   | 8.45           | -                | -           | -       | -                                        |
| M2    | Manway 24"                           | 24"    | 150#   | 3.30           | -                | -           | -       | -                                        |

### C-104 - Ago side stripper

| Mark | Service                     | Size | Rating | z from BTL (m) | Flow (m3/h act.) | rho (kg/m3) | v (m/s) | Criterion                                |
|------|-----------------------------|------|--------|----------------|------------------|-------------|---------|------------------------------------------|
| N1   | Ago feed from C-101 tray 32 | 10"  | 150#   | 8.00           | 79               | 636         | 0.4     | gravity draw to side stripper <= 0.6 m/s |
| N2   | Vapour return to C-101      | 6"   | 150#   | 9.20           | 1,382            | 3.97        | 20.6    | vapour return rho.v2 <= 4000 Pa          |
| N3   | Stripping steam             | 3"   | 150#   | 4.05           | 575              | 1.56        | 33.5    | stripping steam <= 35 m/s                |
| N4   | Bottoms to P-111            | 8"   | 150#   | -0.35          | 71               | 649         | 0.6     | pump suction (bottoms) <= 1.0 m/s        |

| Mark  | Service                              | Size   | Rating | z from BTL (m) | Flow (m3/h act.) | rho (kg/m3) | v (m/s) | Criterion |
|-------|--------------------------------------|--------|--------|----------------|------------------|-------------|---------|-----------|
| N5A/B | LT / LG bridle                       | 2 x 2" | 150#   | 0.20           | -                | -           | -       | -         |
| N6    | Vent / steam-out / PSV-1009 (common) | 3"     | 150#   | 9.20           | -                | -           | -       | -         |
| M1    | Manway 24"                           | 24"    | 150#   | 8.45           | -                | -           | -       | -         |
| M2    | Manway 24"                           | 24"    | 150#   | 3.30           | -                | -           | -       | -         |

### C-105 - Naphtha stabiliser (debutaniser)

| Mark  | Service                                     | Size   | Rating | z from BTL (m) | Flow (m3/h act.) | rho (kg/m3) | v (m/s) | Criterion                                             |
|-------|---------------------------------------------|--------|--------|----------------|------------------|-------------|---------|-------------------------------------------------------|
| N1    | Feed (tray 13)                              | 8"     | 300#   | 8.98           | 187              | 614         | 1.6     | pumped liquid return <= 2.5 m/s                       |
| N2    | Overhead vapour to A-106                    | 8"     | 300#   | 18.00          | 856              | 22          | 7.4     | vapour outlet rho.v2 <= 3000 Pa                       |
| N3    | Reflux                                      | 3"     | 300#   | 16.35          | 23               | 530         | 1.3     | reflux / pumped liquid <= 2.0 m/s                     |
| N4    | Reboiler feed to E-116 (kettle)             | 16"    | 300#   | -0.53          | 380              | 554         | 0.9     | gravity draw / pump suction <= 0.9 m/s (self-venting) |
| N5    | Reboiler return from E-116 (kettle)         | 10"    | 300#   | 4.22           | 614              | 166         | 3.4     | reboiler vapour return rho.v2 <= 4000 Pa              |
| N6    | Stabilised naphtha (bottoms) to E-114/C-106 | 12"    | 300#   | 0.30           | 196              | 554         | 0.7     | pump suction (bottoms) <= 1.0 m/s                     |
| N7    | PSV-1005 relief (R)                         | 6"     | 300#   | 18.00          | -                | -           | -       | -                                                     |
| N8A/B | LT / LG bridle                              | 2 x 2" | 300#   | 0.20           | -                | -           | -       | -                                                     |
| N9    | Vent / steam-out                            | 2"     | 300#   | 18.00          | -                | -           | -       | -                                                     |
| M1    | Manway 24"                                  | 24"    | 300#   | 17.00          | -                | -           | -       | -                                                     |
| M2    | Manway 24"                                  | 24"    | 300#   | 8.98           | -                | -           | -       | -                                                     |
| M3    | Manway 24"                                  | 24"    | 300#   | 3.52           | -                | -           | -       | -                                                     |

### C-106 - Naphtha splitter

| Mark  | Service                                   | Size   | Rating | z from BTL (m) | Flow (m3/h act.) | rho (kg/m3) | v (m/s) | Criterion                                             |
|-------|-------------------------------------------|--------|--------|----------------|------------------|-------------|---------|-------------------------------------------------------|
| N1    | Feed (tray 21)                            | 8"     | 150#   | 15.60          | 196              | 554         | 1.7     | pumped liquid return <= 2.5 m/s                       |
| N2    | Overhead vapour to A-107                  | 20"    | 150#   | 29.50          | 14,876           | 5.72        | 22.0    | vapour outlet rho.v2 <= 3000 Pa                       |
| N3    | Reflux                                    | 6"     | 150#   | 27.85          | 90               | 640         | 1.3     | reflux / pumped liquid <= 2.0 m/s                     |
| N4    | Reboiler feed to E-117 (thermosyphon)     | 16"    | 150#   | -0.78          | 370              | 624         | 0.9     | gravity draw / pump suction <= 0.9 m/s (self-venting) |
| N5    | Reboiler return from E-117 (thermosyphon) | 20"    | 150#   | 4.13           | 11,154           | 6.89        | 21.5    | reboiler vapour return rho.v2 <= 4000 Pa              |
| N6    | Bottoms to P-117A/B                       | 10"    | 150#   | 0.30           | 130              | 624         | 0.7     | pump suction (bottoms) <= 1.0 m/s                     |
| N7    | PSV-1007 relief (T)                       | 8"     | 150#   | 29.50          | -                | -           | -       | -                                                     |
| N8A/B | LT / LG bridle                            | 2 x 2" | 150#   | 0.20           | -                | -           | -       | -                                                     |
| N9    | Vent / steam-out                          | 2"     | 150#   | 29.50          | -                | -           | -       | -                                                     |
| M1    | Manway 24"                                | 24"    | 150#   | 28.50          | -                | -           | -       | -                                                     |
| M2    | Manway 24"                                | 24"    | 150#   | 21.70          | -                | -           | -       | -                                                     |
| M3    | Manway 24"                                | 24"    | 150#   | 15.61          | -                | -           | -       | -                                                     |
| M4    | Manway 24"                                | 24"    | 150#   | 9.51           | -                | -           | -       | -                                                     |
| M5    | Manway 24"                                | 24"    | 150#   | 3.43           | -                | -           | -       | -                                                     |

## 12. Shell & tube exchangers and air coolers - mechanical summary

| Tag   | TEMA size     | Shells | m2/shell | Tubes/shell | Tube OD/L (mm) | Des. P sh/tu (barg) | Des. T | Shell matl. | Shell t | Channel t | Empty t/shell | Hydro t/shell |
|-------|---------------|--------|----------|-------------|----------------|---------------------|--------|-------------|---------|-----------|---------------|---------------|
| E-101 | 1200-6096 AES | 1      | 359      | 756         | 25.4 / 6096    | 30 / 20             | 185    | SA-516-70   | 20      | 14        | 18.9          | 27.2          |
| E-102 | 1300-6096 AES | 1      | 438      | 923         | 25.4 / 6096    | 30 / 20             | 240    | SA-516-70   | 20      | 16        | 22.4          | 32.0          |
| E-103 | 1250-6096 AES | 2      | 402      | 848         | 25.4 / 6096    | 30 / 20             | 255    | SA-516-70   | 20      | 14        | 20.7          | 29.6          |
| E-104 | 1200-6096 AES | 1      | 378      | 797         | 25.4 / 6096    | 30 / 20             | 235    | SA-516-70   | 20      | 14        | 19.5          | 27.7          |
| E-105 | 1300-6096 AES | 1      | 431      | 910         | 25.4 / 6096    | 30 / 20             | 285    | SA-516-70   | 22      | 16        | 22.7          | 32.3          |

| Tag   | TEMA size     | Shells | m2/shell | Tubes/shell | Tube OD/L (mm) | Des. P sh/tu (barg) | Des. T | Shell matl. | Shell t | Channel t | Empty t/shell | Hydro t/shell |
|-------|---------------|--------|----------|-------------|----------------|---------------------|--------|-------------|---------|-----------|---------------|---------------|
| E-106 | 1350-6096 AES | 2      | 500      | 1055        | 25.4 / 6096    | 30 / 20             | 270    | SA-516-70   | 22      | 16        | 25.2          | 35.6          |
| E-107 | 1150-6096 AES | 1      | 324      | 683         | 25.4 / 6096    | 30 / 20             | 325    | SA-516-70   | 22      | 18        | 18.2          | 25.8          |
| E-108 | 1400-6096 AES | 4      | 549      | 1157        | 25.4 / 6096    | 30 / 20             | 345    | SA-387-11-2 | 25      | 18        | 28.2          | 39.3          |
| E-109 | 1000-6096 AES | 1      | 263      | 555         | 25.4 / 6096    | 30 / 20             | 365    | SA-387-11-2 | 20      | 16        | 14.5          | 20.2          |
| E-110 | 1300-6096 AES | 2      | 443      | 934         | 25.4 / 6096    | 30 / 20             | 345    | SA-387-11-2 | 22      | 18        | 23.2          | 32.9          |
| E-111 | 1400-6096 AES | 2      | 547      | 1154        | 25.4 / 6096    | 30 / 20             | 395    | SA-387-11-2 | 25      | 18        | 28.1          | 39.3          |
| E-113 | 1300-6096 AKT | 1      | 223      | 627         | 19.05 / 6096   | 13 / 20             | 290    | SA-516-70   | 14      | 12        | 10.3          | 20.2          |
| E-201 | 1950-6096 AKT | 1      | 648      | 1820        | 19.05 / 6096   | 6 / 20              | 310    | SA-516-70   | 16      | 14        | 24.2          | 46.4          |
| E-114 | 750-6096 AES  | 1      | 117      | 248         | 25.4 / 6096    | 20 / 20             | 235    | SA-516-70   | 10      | 10        | 6.4           | 9.7           |
| E-115 | 900-6096 AEU  | 2      | 370      | 1040        | 19.05 / 6096   | 3.5 / 7             | 90     | SA-516-70   | 12      | 12        | 9.0           | 13.5          |
| E-116 | 1200-6096 BKT | 1      | 190      | 535         | 19.05 / 6096   | 15 / 45             | 285    | SA-516-70   | 14      | 20        | 9.5           | 18.0          |
| E-117 | 750-6096 BXM  | 1      | 225      | 632         | 19.05 / 6096   | 5 / 13              | 215    | SA-516-70   | 10      | 10        | 8.0           | 11.2          |
| E-118 | 600-6096 AEL  | 1      | 84       | 177         | 25.4 / 6096    | 20 / 20             | 160    | SA-516-70   | 10      | 10        | 4.7           | 6.8           |
| E-202 | 1100-6096 AXS | 1      | 549      | 1543        | 19.05 / 6096   | 3.5 / 7 (FV)        | 100    | SA-516-70   | 14      | 14        | 18.6          | 25.4          |
| E-203 | 500-6096 AXS  | 1      | 79       | 222         | 19.05 / 6096   | 3.5 / 7 (FV)        | 100    | SA-516-70   | 10      | 10        | 3.5           | 5.0           |
| E-204 | 400-4877 AXS  | 1      | 30       | 107         | 19.05 / 4877   | 3.5 / 7 (FV)        | 100    | SA-516-70   | 10      | 10        | 1.9           | 2.7           |

$Shell\ t = \max(UG-27\ with\ E = 0.85 + CA, \text{TEMA } R\ minimum)$ . Bundle diameter from tube-count correlation (square pitch 1.25 OD for fouling crude; triangular for clean condensers).

| Tag   | Service                       | Bare area (m2) | Bays (data) | Rows req. | Bays @ 6 rows | Fans | Fan D (m) | Motor kW | Empty (t) |
|-------|-------------------------------|----------------|-------------|-----------|---------------|------|-----------|----------|-----------|
| A-101 | Atm. overhead condenser       | 1,667          | 2           | 10        | 4             | 4    | 4.5       | 90       | 102.7     |
| A-103 | Kero cooler                   | 184            | 1           | 3         | 1             | 2    | 4.5       | 11       | 23.8      |
| A-104 | Diesel cooler                 | 430            | 1           | 6         | 1             | 2    | 4.5       | 45       | 34.2      |
| A-105 | Ago cooler                    | 287            | 1           | 4         | 1             | 2    | 4.5       | 30       | 28.2      |
| A-106 | Stabiliser overhead condenser | 214            | 1           | 3         | 1             | 2    | 4.5       | 11       | 25.1      |
| A-107 | Splitter overhead condenser   | 881            | 1           | 11        | 2             | 2    | 4.5       | 45       | 53.4      |
| A-108 | Heavy naphtha product cooler  | 401            | 1           | 5         | 1             | 2    | 4.5       | 30       | 33.0      |
| A-201 | Lvgo cooler                   | 397            | 1           | 5         | 1             | 2    | 4.5       | 15       | 32.8      |
| A-202 | Hvgo Product cooler           | 403            | 1           | 5         | 1             | 2    | 4.5       | 37       | 33.1      |

## 13. Process-data issues and holds

| #  | Item      | Issue / recommendation                                                                                                                                                                                                                                                                                                                                                                             |
|----|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | C-201     | Top section ID 2.6 m is sized on vapour leaving the top of bed 1 (10 t/h). Vapour entering the bottom of bed 1 is ~ vapour leaving bed 2 (87 t/h), which needs ID ~4.8 m at Cs 0.11. Recommend process re-rate bed 1 at its bottom; GA drawn per equipment.json (hold).                                                                                                                            |
| 2  | H-101     | des_T 455 C in equipment.json is below the API 530 design metal temperature 510 C (estimated EOR TMT 494 C + 15 C). Use 510 C for coil design / datasheet.                                                                                                                                                                                                                                         |
| 3  | H-201     | equipment.json L = 14.0 m cannot house 18.3 m horizontal radiant tubes; cabin incl. header boxes is 21.9 m long. Plot reservation must be >= 24 m (incl. tube-pulling at one end handled by crane).                                                                                                                                                                                                |
| 4  | H-201     | des_T 490 C in equipment.json is below the API 530 design metal temperature 575 C (estimated EOR TMT 559 C + 15 C). Use 575 C for coil design / datasheet.                                                                                                                                                                                                                                         |
| 5  | H-101     | Radiant fraction 0.65 (basis.DESIGN) gives an estimated bridgwall temperature of 1044 C with APH; API 560 practice <= ~900-950 C. Suggest radiant fraction ~0.70 (more radiant surface) - vendor to optimise.                                                                                                                                                                                      |
| 6  | H-201     | 88 % efficiency is not achievable with process convection alone: charge enters at 349 C, so flue gas cannot be cooled below ~399 C (max. process convection 3.7 MW vs 5.1 MW assumed). ~1.4 MW must go to a utility coil (steam superheat / BFW) or an APH - none exists in equipment.json. Also vacuum-heater outlet tubes normally step up in size (6"->8"->10") - all tubes are 6" in the data. |
| 7  | P-112A/B  | Rated dP 12 bar gives discharge ~13.9 barg but H&MB; stream 19 (P-112 -> H-201) is at 19.0 barg: dP should be ~17 bar (head ~246 m).                                                                                                                                                                                                                                                               |
| 8  | P-206A/B  | Rated flow 0.65 m3/h at 60 m is outside the centrifugal (API 610) range - use a positive-displacement / metering pump (API 674/675).                                                                                                                                                                                                                                                               |
| 9  | P-118     | Single pump without spare (desalter mud-wash/recycle) - confirm intermittent duty.                                                                                                                                                                                                                                                                                                                 |
| 10 | Stream 24 | HVGO product 'frm' is E-202 (ejector intercondenser); should be A-202 (HVGO product cooler).                                                                                                                                                                                                                                                                                                       |
| 11 | Stream 20 | Destination 'C-102A-C' - strippers are tagged C-102/C-103/C-104.                                                                                                                                                                                                                                                                                                                                   |
| 12 | A-101     | 1667 m2 bare in 2 bay(s) of 6 x 12 m needs 10 tube rows (>8, not practical); 4 bays at 6 rows required. Plot space / fan count to be revised.                                                                                                                                                                                                                                                      |
| 13 | A-107     | 881 m2 bare in 1 bay(s) of 6 x 12 m needs 11 tube rows (>8, not practical); 2 bays at 6 rows required. Plot space / fan count to be revised.                                                                                                                                                                                                                                                       |
| 14 | A-202     | Air temperature rise 35->85 C is unrealistically high (typ. 15-25 C); air flow and bundle size will increase.                                                                                                                                                                                                                                                                                      |

| #  | Item      | Issue / recommendation                                                                                                                                                                                                                                                                |
|----|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15 | X-104     | Area 'CDU' but service is VDU overhead (area 200) - sizing.py tests "'10" in tag'.                                                                                                                                                                                                    |
| 16 | D-102     | moc states 6 mm CA but ca_mm = 3; mechanical design uses 6 mm.                                                                                                                                                                                                                        |
| 17 | E-116     | Tube-side HP steam header is superheated (400 C) but des_T = 285 C - design temperature must cover the steam supply (400 C) unless a desuperheater is added.                                                                                                                          |
| 18 | E-117     | Tube-side MP steam header is superheated (250 C) but des_T = 215 C - design temperature must cover the steam supply (250 C) unless a desuperheater is added.                                                                                                                          |
| 19 | PSV-1001  | 4 x T orifices for 'Reflux failure / blocked OH (total OH vapour)' (216 t/h, set 2.4 barg vs MAWP 3.5 barg). Raising set to MAWP and crediting unaffected pumparound duty (API 521 4.4.3) should reduce this.                                                                         |
| 20 | PSV-1009  | One PSV protects C-102/103/104: requires no isolation between the vessels (locked-open valves) - confirm in P&ID.;                                                                                                                                                                    |
| 21 | E-1xx/2xx | Shell IDs from tube-count/bundle correlation exceed equipment.json 'D' for: E-113 1.30 vs 0.9, E-201 1.95 vs 1.2, E-116 1.20 vs 0.9 m (kettles include the enlarged shell).                                                                                                           |
| 22 | C-101     | equipment.json weight_t 727 t (rule of thumb) vs calculated empty 353 t / operating 438 t - use data/mech.json.                                                                                                                                                                       |
| 23 | C-201     | equipment.json weight_t 665 t (rule of thumb) vs calculated empty 294 t / operating 337 t - use data/mech.json.                                                                                                                                                                       |
| 24 | Design P  | Only C-201 (and ejector condensers) carry FV. Columns C-101..C-106 and drums D-101A, D-101B, D-102, D-103, D-104, D-106, D-201 are not full-vacuum capable as designed; refinery practice is FV for steam-out (or written steam-out procedure + vacuum breakers). Process to confirm. |
| 25 | C-101     | Stripping-section ID 4.14 m (= 0.6 x main ID) is not rounded; size string says 4.1 m. Hydraulically 2.5 m suffices; mechanical uses 4.14 m as given.                                                                                                                                  |

## 14. Summary of results

| Tag    | Service                                | Size (m)           | Material      | Shell t (mm) | Head t (mm) | Skirt (m / mm) | Empty (t) | Operating (t) | Hydrotest (t) |
|--------|----------------------------------------|--------------------|---------------|--------------|-------------|----------------|-----------|---------------|---------------|
| C-101  | Atmospheric crude fractionator         | 6.9 / 4.14 x 40.5  | SA-516-70     | 16-20        | 16/12       | 6.5 / 20       | 353       | 438           | 1,629         |
| C-102  | Kero side stripper                     | 1.7 x 9.2          | SA-516-70     | 8            | 10/10       | 4 / 8          | 20        | 23            | 41            |
| C-103  | Diesel side stripper                   | 1.9 x 9.2          | SA-516-70     | 12           | 12/12       | 4.5 / 8        | 27        | 30            | 53            |
| C-104  | Ago side stripper                      | 1.4 x 9.2          | SA-516-70     | 10           | 12/12       | 3.5 / 8        | 18        | 20            | 32            |
| C-105  | Naphtha stabiliser (debutaniser)       | 2.1 x 18           | SA-516-70     | 14           | 16/16       | 6 / 8          | 48        | 54            | 111           |
| C-106  | Naphtha splitter                       | 3.1 x 29.5         | SA-516-70     | 12           | 12/12       | 7.5 / 12       | 109       | 130           | 334           |
| C-201  | Vacuum column (wet, packed)            | 6.6 / 3 / 2.6 x 36 | SA-516-70     | 18-25        | 18/18       | 6 / 18         | 294       | 337           | 1,043         |
| D-101A | Electrostatic desalter, 1st stage      | 3.8 x 30           | SA-516-70     | 28           | 28          | saddles        | 137       | 414           | 487           |
| D-101B | Electrostatic desalter, 2nd stage      | 3.8 x 30           | SA-516-70     | 28           | 28          | saddles        | 137       | 414           | 487           |
| D-102  | Atm. overhead reflux drum (3-phase)    | 3.4 x 12           | SA-516-70     | 12           | 12          | saddles        | 19        | 63            | 144           |
| D-103  | Fuel gas knock-out drum                | 0.9 x 2.5          | SA-516-70     | 8            | 8           | legs           | 1         | 2             | 3             |
| D-104  | Flare knock-out drum (unit)            | 2.1 x 6.5          | SA-516-70     | 8            | 8           | saddles        | 5         | 14            | 29            |
| D-105  | Stabiliser reflux drum                 | 1.7 x 5.5          | SA-516-70     | 16           | 16          | saddles        | 6         | 11            | 20            |
| D-106  | Splitter reflux drum                   | 2.7 x 8.5          | SA-516-70     | 10           | 10          | saddles        | 9         | 28            | 63            |
| D-201  | Ejector hotwell / sour water separator | 1.3 x 5.5          | SA-516-70     | 8            | 8           | saddles        | 3         | 6             | 10            |
| D-202  | Vacuum off-gas knock-out drum          | 0.6 x 2            | SA-516-70     | 8            | 8           | legs           | 1         | 1             | 1             |
| H-101  | Atmospheric crude charge heater        | 21.9 x 12.7 x 38   | A335 P9 tubes | 10.97 (tube) | -           | -              | 1,033     | 1,059         | 905           |
| H-201  | Vacuum heater                          | 21.9 x 5.4 x 32    | A335 P9 tubes | 10.97 (tube) | -           | -              | 328       | 337           | 347           |